

Determining the Effective Temperature Range That *Pestalotiopsis Microspora* Can Degrade PET and Polyurethane For Bioremediation Purposes

Introduction

With most plastics being made of hazardous, nonrenewable petroleum-based chemicals, plastic pollution has become a major concern in today's modern world. Most of these plastics are unrecyclable or remain unrecycled with only about 9% of plastics in the world being recycled (OECD, 2022). This leaves a large portion of the world's plastic (91%) to be either landfilled, incinerated or mismanaged, all of which negatively effect the environment. These plastics consist of commonly used plastics such as PVC and PET and more uncommonly used ones such as PLA, PBT and PU. These plastics tend to have varying properties and uses, with some such as PET being recyclable. However, recyclable plastics such as PET are still often left unrecycled and often end up in the same areas as unrecyclable plastics.

Background

Polyesters are a family of plastics that consist of repeating subunits linked by ester bonds (CO-O). These plastics have a wide range of application and properties being used in most everyday products such as bottles, textiles and packaging. Polyester Polyurethane (PUR) was found to be degraded by the fungus *Pestalotiopsis Microspora*, with this degradation being possible due to the fungus producing an esterase, a type of enzyme that specializes in breaking ester bonds (Russel, 2011). Additionally, the fungus produces other enzymes such as cysteine hydrolase and serine hydrolase to aid in degradation (Kothawale, 2023). Enzymes often work most effectively at specific temperature ranges. Additionally, most microorganisms have a preferred temperature range where they can thrive in.

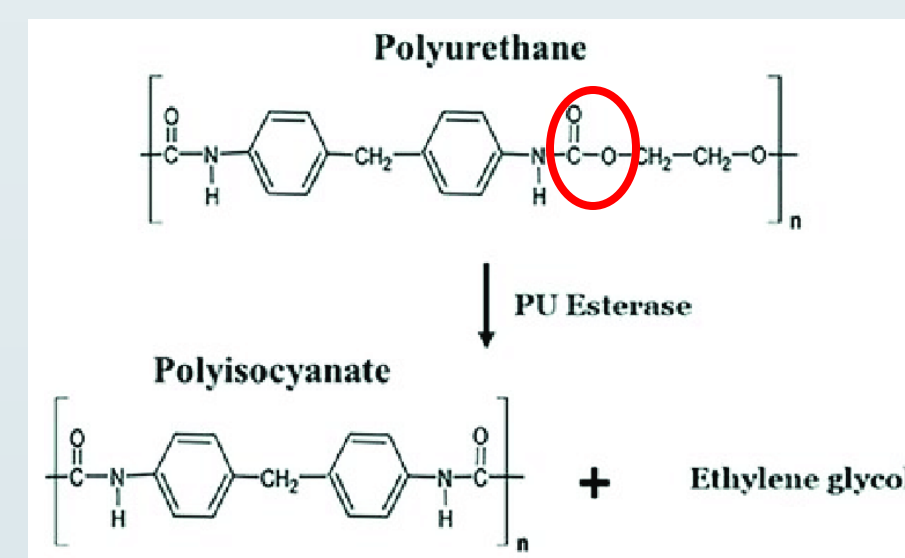


Figure 1. Breakdown of PUR by esterase
<https://www.researchgate.net/profile/Natalia-Tkachuk/publication/354041689/enr/f142-11993845692254403/figure/fig1/figure-fig1/2023/01/20/2023012020/Microbial-degradation-pathway-of-polyurethane-56-TCA-cycle-the-cycle-of-tricarboxylic.png>

Purpose

Up to this point, some research has been done on the bioremediation capabilities of *Pestalotiopsis Microspora*. With most focusing on the fungus's ability to degrade Polyurethane. The purpose of this project is to determine the temperature range tat *Pestalotiopsis Microspora* can degrade Polyester Polyurethane and PET. The data collected would help to determine the effectiveness of *Pestalotiopsis Microspora* as a possible biodegradation candidate. As well as the temperature that *Pestalotiopsis Microspora* most effectively degrades each of the plastics..

Methodology

Stage 1: Creating Stock Culture

1. Obtain culture of *Pestalotiopsis Microspora*
2. Create PDA and inoculate
 - Create PDA to serves as an agar medium that *Pestalotiopsis Microspora* is known to grow on
 - Inoculate using sterile scalpel and tissue sample from culture. Do so underneath a laminar flow hood
3. Incubate inoculated agar plates
 - Plates are incubated in an incubator at 27 degrees Celsius, which is known incubation

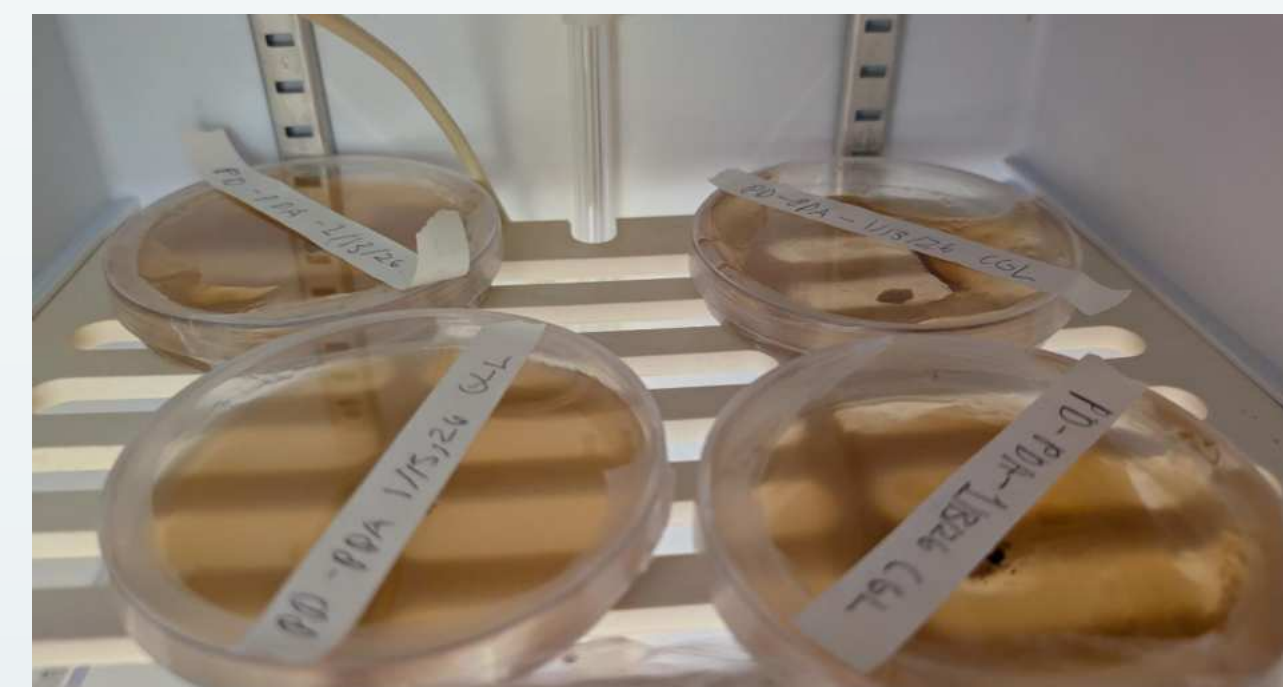


Figure 2. Inoculated PDA agar plates being incubated at 27 degrees Celsius

Stage 3: Combining Plastic Dispersions and Agar

1. Use magnetic stir bar to vigorously stir agar mixture
2. Use micropipette to pipette plastic dispersion into agar



Figure 6. PET base agar mixture being vigorously stirred prior to addition of dissolved PET

Stage 5: Inoculating Agar Plates

1. Agar plates would be inoculated using sterile scalpel and tissue sample from stock culture of *Pestalotiopsis Microspora*
2. One of each type of agar plate would not be inoculated to act as a control
3. Agar plates would be incubated between 18-28 degrees Celsius



Figure 10. Plastic infused agar plates inoculated with *Pestalotiopsis Microspora*

Stage 2: Creating Base Agar Mixture

1. Create base agar mixture for PET
 - Combine 3.1g of agar, 20ml of 10x PBS, and 0.02g of *N*-Lauroylsarcosine sodium salt
 - Add distilled water until volume equal to 200ml
 - Autoclave mixture
2. Create base agar mixture for PUR
 - Mix 1.5g of agar with distilled water for total volume of 100ml
 - Autoclave mixture
3. Obtain samples of base agar mixture



Figure 4 & 5. Base agar mixtures within beakers and poured onto aluminum to act as samples for FTIR

Stage 4: Creating Agar Plates

1. Pour plastic infused agar into agar plates
2. Label agar plates and let cool
3. Use remaining plastic infused agar to create samples



Figure 7, 8 & 9. Plastic infused agar plates (PET and PUR) with PET agar sample

Stage 6: Collecting Data

1. Agar plates would be dried after 2 weeks of being incubated
2. FTIR scans would be done on agar plates
3. FTIR scans would be done on samples taken previously

Stage 7: Data Processing

1. FTIR scans of samples, controls, and inoculated agar would be compared.
2. Disappearance of absorbance peaks at 1700, 1200 and 1100 would indicate broken ester bonds.
3. Amount of degradation occurred at each temperature would be compared.

Results

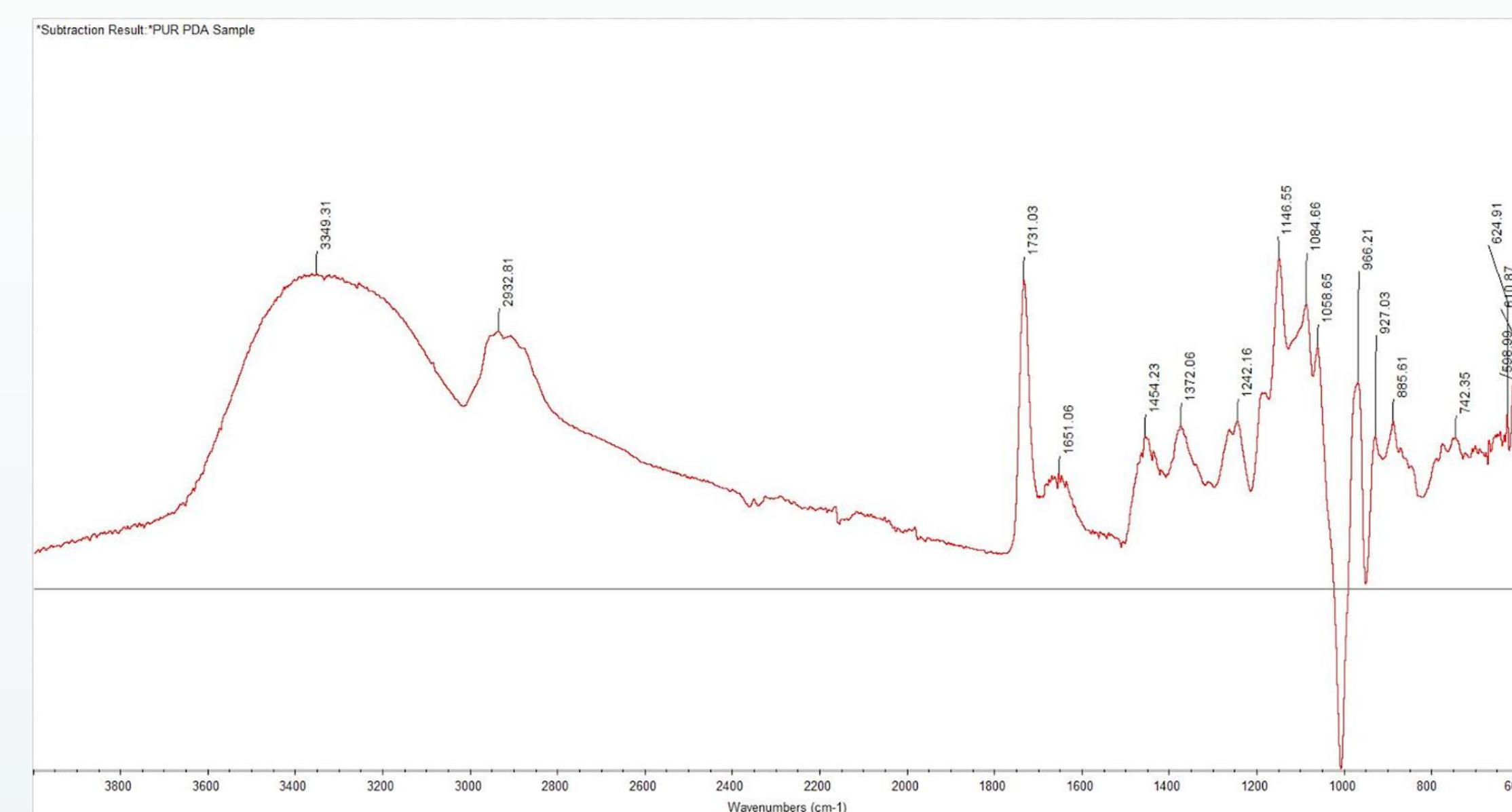


Figure 11. FTIR subtraction of PUR agar and base agar. Demonstrates absorbance peaks

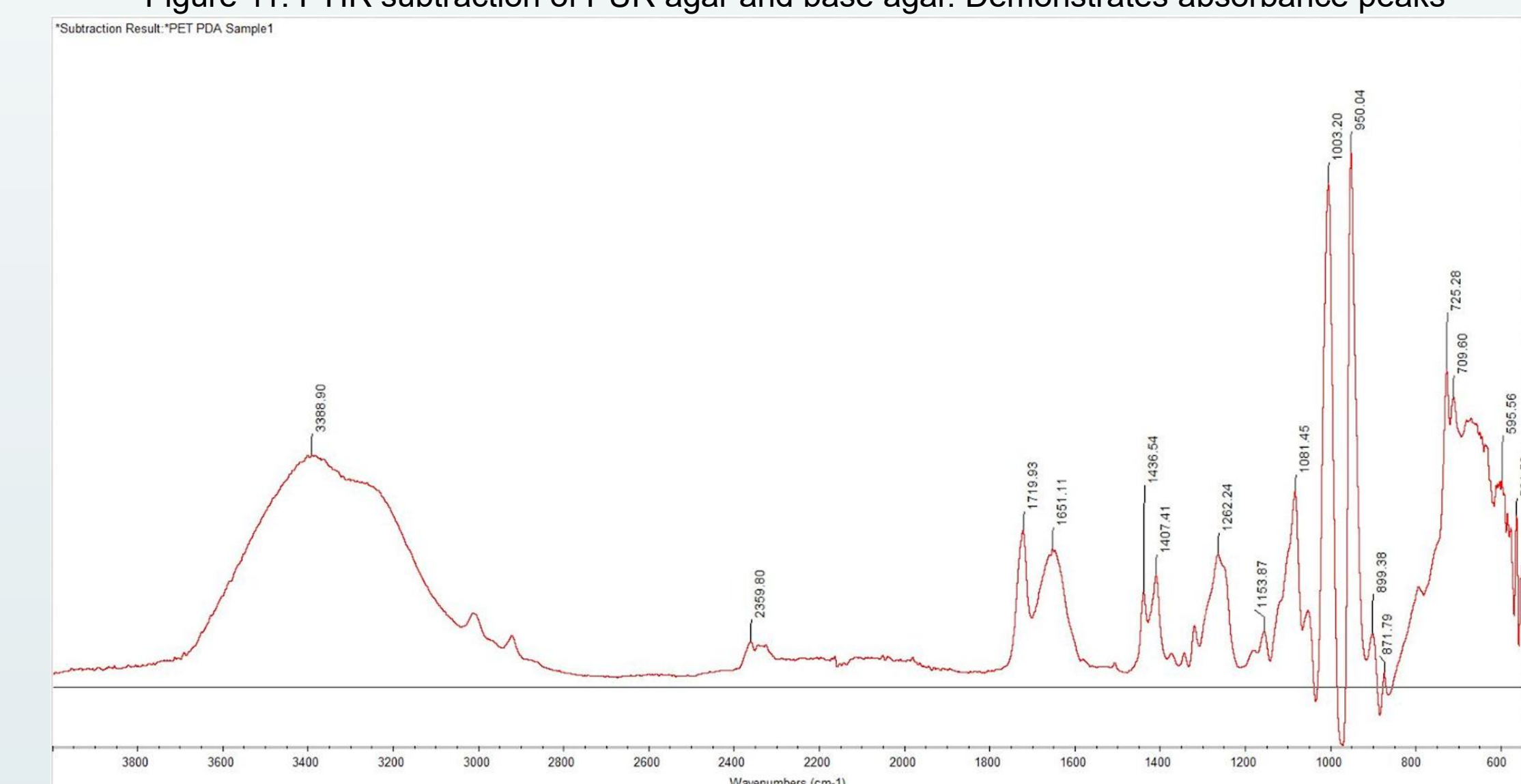


Figure 12. FTIR subtraction of PET agar and base agar. Demonstrates absorbance peaks
Note: The data from the degradation experiment has not yet been recorded as the experiment could not be conducted due to issues encountered with receiving the organism

Conclusion

As of now no definitive conclusions can be drawn as the trials have not yet been conducted, due to complications. Progress has been made in creating and verifying a procedure that can effectively disperse PET and PUR within agar for use in biodegradation studies. Additionally, the project confirms that *Pestalotiopsis Microspora* is not a beginner friendly fungus species and should be worked with by people who have access to laboratory equipment and are familiar with mycology.

Acknowledgements

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